Sharath Jotawar

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Location: Singapore
Visa Status: On Employment Pass
Nationality: Indian

Professional Summary

- 11+ years of experience building real-time embedded systems combining multi-camera vision and radar sensor fusion.
- Specialized in 3D perception, multi-view geometry, object tracking, and robotic vision systems.
- Proven track record of delivering production-grade systems for multi-sensor calibration, trajectory estimation, and autonomous robotic platforms.
- Strong background in C++, Python, and deep learning with expertise in OpenCV, ROS, and real-time system design.

Key Highlights

- Built real-time multi-sensor (camera + radar) baseball tracking system operating at 125 FPS.
- Designed multi-camera calibration and sensor fusion pipelines for accurate 3D reconstruction
- Developed autonomous robotic system integrating perception and planning for pick and place application
    - 3rd place – Amazon Robotics Challenge (ARC 2017), RoboCup Japan

Skill Sets

- **Programming**: C++, Python, HTML, CSS, JavaScript
- **Computer Vision**: Multi-view geometry, camera calibration, object detection, tracking, 3D reconstruction
- **Machine Learning**: Deep Learning (TensorFlow, Keras), CNN models for detection and segmentation
- **Robotics**: ROS, MoveIt, Gazebo, motion planning, autonomous systems, FlexBE
- **3D Processing**: PCL, point cloud processing, pose estimation
- **Systems**: Real-time systems, multi-threading, Linux
- **Tools**: OpenCV, NumPy, Pandas, Git

Work Experience

Rapsodo Pte Ltd, Singapore as Staff Computer Vision Engineer **Dec 2020 to Present**

- Designed and deployed real-time multi-sensor system combining multi-camera vision and radar for accurate baseball trajectory estimation.
- Developed multi-camera calibration pipelines (intrinsic & extrinsic) improving 3D reconstruction accuracy and system robustness.
- Built sensor fusion algorithms integrating radar and vision data to enhance tracking accuracy under high-speed motion conditions.
- Implemented object detection and tracking pipelines for detection of fast-moving baseballs using computer vision techniques optimized for real-time performance. Achieved processing speed 125 FPS with 99% recall.
- Contributed to production release of the product Rapsodo-Pro 2.0 used in baseball and softball analytics.

Transforma Robotics Pte Ltd, Singapore as Software Engineer **Mar 2018 to Dec 2020**

- Developed Mask R-CNN–based perception system for semantic segmentation in indoor environments.
- Designed motion and task planning system for autonomous painting robot, enabling fully automated operation.
- Built end-to-end robotic pipeline integrating perception, planning, and execution for industrial deployment.

TCS Innovation Labs, Bangalore, India as Researcher **Aug 2014 to Mar 2018**

- Developed real-time object detection systems using Faster R-CNN for cluttered environments.
- Designed 3D perception pipelines for robotic grasping using point cloud processing (PCL) Demo Video.
- Implemented object pose estimation using SUPER4PCS for robust grasp planning Demo Video.

Continental Automotive Components India Pvt Ltd as Graduate Engineer Trainee **Aug 2010 to May '11**

Responsibilities: Conducting verification of circuit design of different modules in prototype Engine Control Unit.

Achievements

- **Amazon Robotics Challenge** (RoboCup-2017, Japan) as software lead in Team IITK-TCS
  - 3rd place - pick task
  - 4th place overall (16 international teams).
- Publications in robotic pick-and-place systems and motion planning (3 Papers)
  - Design and development of an automated robotic pick & stow system for an e-commerce warehouse.
  - Methods for improving performance of robotic pick and place system for ARC 2017.
  - Motion planning for an automated pick and place robot in a retail warehouse.

Academic Background

M Tech. in Electronics & Electrical Engineering with Specialization in Signal Processing **Institute: Indian Institute of Technology Guwahati (IIT Guwahati), India** **Thesis**: Improving Sparse Representation using NN-LASSO for Robust Automatic Speech Recognition. Implemented using GMM-HMM based HTK tool kit.	Yr: 2012-14 CPI: 8.34
B.E. in Electronics & Communication Engineering **Institute: BMS College of Engineering, Bangalore, India.**	Yr: 2006-10 Avg: 71.9 %